CSC 402 — MILESTONE 3-A

Database Design — Submission Template

Due: Week 12 • 5 marks • One submission per team (PDF)

How to use this template

1. Fill every yellow cell. Replace the grey example rows with your own data and delete the grey hint lines.
2. Be brief and exact: short phrases, real names, real fields — no general explanations.
3. Design the database for your own app's features from Milestone 2. Copy a table as many times as you need.
4. Export as PDF and name it Team##_M3A_Database.pdf. Commit the same files to /docs/database in GitHub.

How you will be graded (100 points = 5 marks)

Criterion	Section	Pts
ER diagram: all entities, keys and relationships with cardinality	Section 2	**30**
Relational schema: fields, types, keys and constraints	Section 3, 4	**30**
Screen-to-data map: every screen has its data	Section 5	**20**
Room plan: entities, DAOs and key queries	Section 6	**10**
GitHub evidence and AI Use Log	Section 7, 8	**10**

Cover Sheet

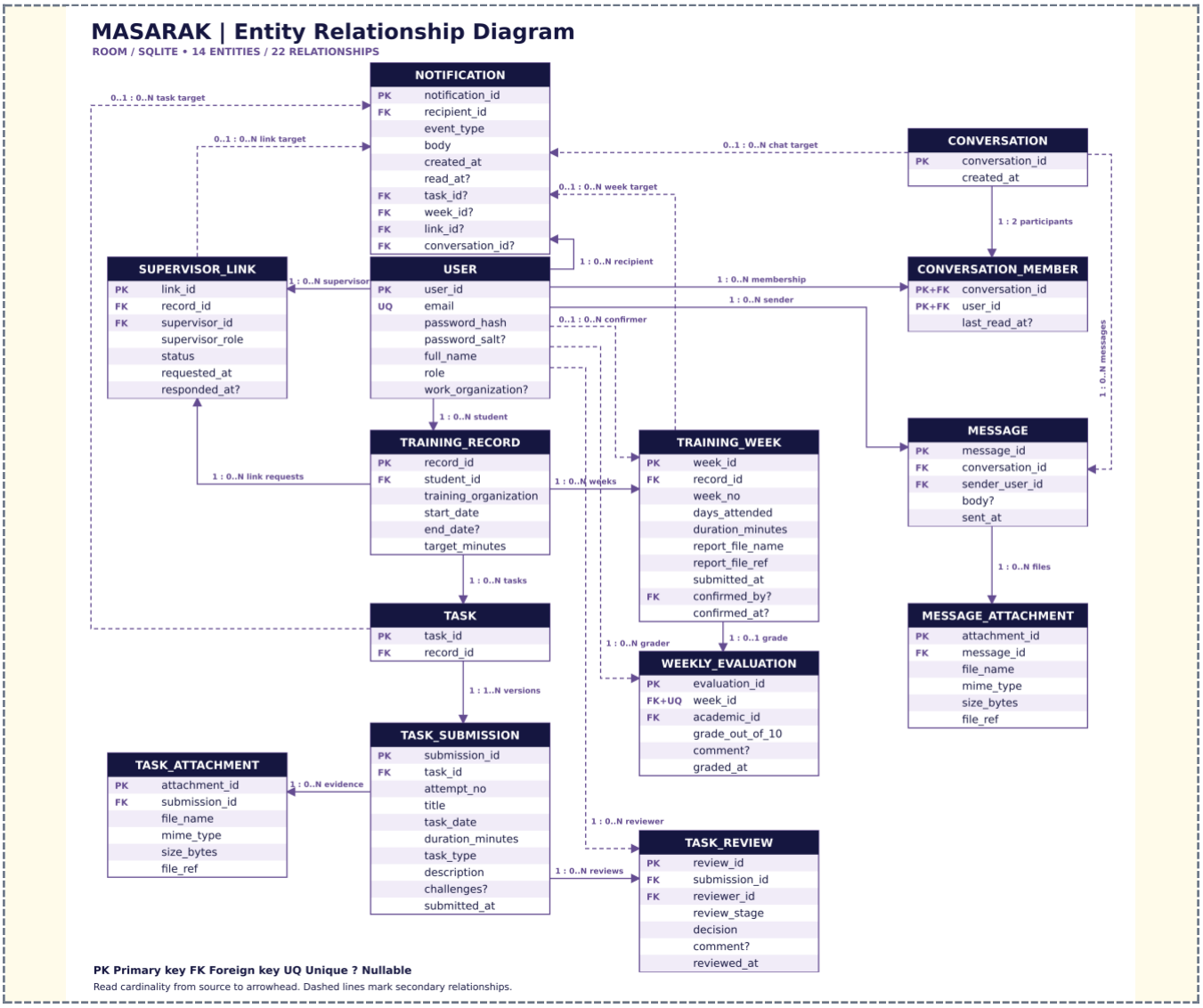
Team name / number	Group 01	App name	MASARAK | مسارك
GitHub repository link	https://github.com/janashammar1/MASARAK-Mobile-App	Database type	Room (SQLite)

#	Student name	Student ID	Role in this milestone
1	Jana Al-Shammary	2240006642	Team leader
2	Maha AlJoaib	2240006849	Team member
3	Alya Shehab	2240002062	Team member
4	Raghad Alqahtani	2240001622	Team member
5	Aisha Alhuzaym	2240002568	Team member
6	Fatima Alomran	2240006720	Team member
7	Rawan Muslim	2240001473	Team member

1. Entities at a Glance

Entity (table name)	What it stores	Related to
USER	Accounts, login credentials and roles	Referenced by other entities
TRAINING_RECORD	Student placements and training targets	USER
SUPERVISOR_LINK	Requests linking supervisors to placements	TRAINING_RECORD, USER
TRAINING_WEEK	Weekly attendance totals, reports and confirmations	TRAINING_RECORD, USER
WEEKLY_EVALUATION	Academic grades and comments for training weeks	TRAINING_WEEK, USER
TASK	Stable task identities within a placement	TRAINING_RECORD
TASK_SUBMISSION	Versioned task submissions	TASK
TASK_ATTACHMENT	Files submitted as task evidence	TASK_SUBMISSION
TASK_REVIEW	Field and academic review decisions	TASK_SUBMISSION, USER
CONVERSATION	Chat threads	Referenced by other entities
CONVERSATION_MEMBER	Users who belong to each chat thread	CONVERSATION, USER
MESSAGE	Messages sent within chat threads	CONVERSATION, USER
MESSAGE_ATTACHMENT	Files attached to chat messages	MESSAGE
NOTIFICATION	User alerts with optional navigation targets	USER, TASK, TRAINING_WEEK, SUPERVISOR_LINK, CONVERSATION

2. ER Diagram


MASARAK | Entity Relationship Diagram
ROOM / SQLITE • 14 ENTITIES / 22 RELATIONSHIPS
NOTIFICATION
PK notification_id
FK recipient_id
event_type
body
created_at
read_at?
FK task_id?
FK week_id?
FK link_id?
FK conversation_id?
CONVERSATION
PK conversation_id
created_at
SUPERVISOR_LINK
PK link_id
FK record_id
FK supervisor_id
supervisor_role
status
requested_at
responded_at?
USER
PK user_id
UQ email
password_hash
password_salt?
full_name
role
work_organization?
CONVERSATION_MEMBER
PK+FK conversation_id
PK+FK user_id
last_read_at?
TRAINING_RECORD
PK record_id
FK student_id
training_organization
start_date
end_date?
target_minutes
TRAINING_WEEK
PK week_id
FK record_id
week_no
days_attended
duration_minutes
report_file_name
report_file_ref
submitted_at
FK confirmed_by?
confirmed_at?
MESSAGE
PK message_id
FK conversation_id
FK sender_user_id
body?
sent_at
TASK
PK task_id
FK record_id
WEEKLY_EVALUATION
PK evaluation_id
FK+UQ week_id
FK academic_id
grade_out_of_10
comment?
graded_at
MESSAGE_ATTACHMENT
PK attachment_id
FK message_id
file_name
mime_type
size_bytes
file_ref
TASK_SUBMISSION
PK submission_id
FK task_id
attempt_no
title
task_date
duration_minutes
task_type
description
challenges?
submitted_at
TASK_ATTACHMENT
PK attachment_id
FK submission_id
file_name
mime_type
size_bytes
file_ref
TASK_REVIEW
PK review_id
FK submission_id
FK reviewer_id
review_stage
decision
comment?
reviewed_at
0..1 : 0..N task target
0..1 : 0..N link target
0..1 : 0..N chat target
0..1 : 0..N week target
1 : 0..N recipient
1 : 2 participants
1 : 0..N supervisor
1 : 0..N membership
1 : 0..N sender
0..1 : 0..N confirmer
1 : 0..N messages
1 : 0..N student
1 : 0..N link requests
1 : 0..N weeks
1 : 0..N tasks
1 : 0..N files
1 : 0..N grader
1 : 0..1 grade
1 : 1..N versions
1 : 0..N evidence
1 : 0..N reviewer
1 : 0..N reviews
PK Primary key FK Foreign key UQ Unique ? Nullable
Read cardinality from source to arrowhead. Dashed lines mark secondary relationships.

3. Relational Schema & Data Dictionary

The ERD uses singular labels and snake_case fields. The tables below pair each label with its Room table; Kotlin properties use camelCase. Nullable Kotlin properties initialize to null where declared; this is not a SQLite DEFAULT.

USER | Room: users

Field	Type	Key	Constraints	Description
user_id	Long	PK	NOT NULL; auto-generated	Identifier of the user
email	String		NOT NULL; UNIQUE (Room unique index)	Account email used for sign-in
password_hash	String		NOT NULL	Stored password hash used for verification
password_salt	String?		NULL allowed	Optional salt used during verification
full_name	String		NOT NULL	Full name
role	String		NOT NULL	Student, Field Supervisor or Academic Supervisor
work_organization	String?		NULL allowed	Optional user organization

TRAINING_RECORD | Room: training_records

Field	Type	Key	Constraints	Description
record_id	Long	PK	NOT NULL; auto-generated	Identifier of the record
student_id	Long	FK → USER	NOT NULL	Identifier of the student
training_organization	String		NOT NULL	Training organization
start_date	Long		NOT NULL	Start date
end_date	Long?		NULL allowed	End date
target_minutes	Int		NOT NULL	Required training duration in minutes

SUPERVISOR_LINK | Room: supervisor_links

Field	Type	Key	Constraints	Description
link_id	Long	PK	NOT NULL; auto-generated	Identifier of the link
record_id	Long	FK → TRAINING_RECORD	NOT NULL	Identifier of the record
supervisor_id	Long	FK → USER	NOT NULL	Identifier of the supervisor
supervisor_role	String		NOT NULL	Role of the supervisor in this link
status	String		NOT NULL	Status
requested_at	Long		NOT NULL	Timestamp for requested

Field	Type	Key	Constraints	Description
responded_at	Long?		NULL allowed	Timestamp for responded

TRAINING_WEEK | Room: training_weeks

Field	Type	Key	Constraints	Description
week_id	Long	PK	NOT NULL; auto-generated	Identifier of the week
record_id	Long	FK → TRAINING_RECORD	NOT NULL	Identifier of the record
week_no	Int		NOT NULL	Week sequence within the training record
days_attended	Int		NOT NULL	Days attended during the week
duration_minutes	Int		NOT NULL	Recorded duration in minutes
report_file_name	String		NOT NULL	Report file name
report_file_ref	String		NOT NULL	Reference to the stored weekly report
submitted_at	Long		NOT NULL	Timestamp for submitted
confirmed_by	Long?	FK → USER	NULL allowed	User who confirmed the training week
confirmed_at	Long?		NULL allowed	Timestamp for confirmed

WEEKLY_EVALUATION | Room: weekly_evaluations

Field	Type	Key	Constraints	Description
evaluation_id	Long	PK	NOT NULL; auto-generated	Identifier of the evaluation
week_id	Long	FK → TRAINING_WEEK	NOT NULL; UNIQUE	Identifier of the week
academic_id	Long	FK → USER	NOT NULL	Identifier of the academic
grade_out_of10	Float		NOT NULL	Academic grade on a 0–10 scale
comment	String?		NULL allowed	Comment
graded_at	Long		NOT NULL	Timestamp for graded

TASK | Room: tasks

Field	Type	Key	Constraints	Description
task_id	Long	PK	NOT NULL; auto-generated	Identifier of the task
record_id	Long	FK → TRAINING_RECORD	NOT NULL	Identifier of the record

TASK_SUBMISSION | Room: task_submissions

Field	Type	Key	Constraints	Description
submission_id	Long	PK	NOT NULL; auto-generated	Identifier of the submission
task_id	Long	FK → TASK	NOT NULL	Identifier of the task

Field	Type	Key	Constraints	Description
attempt_no	Int		NOT NULL	Submission version number for the task
title	String		NOT NULL	Title
task_date	Long		NOT NULL	Task date
duration_minutes	Int		NOT NULL	Recorded duration in minutes
task_type	String		NOT NULL	Task type
description	String		NOT NULL	Description
challenges	String?		NULL allowed	Challenges
submitted_at	Long		NOT NULL	Timestamp for submitted

TASK_ATTACHMENT | Room: task_attachments

Field	Type	Key	Constraints	Description
attachment_id	Long	PK	NOT NULL; auto-generated	Identifier of the attachment
submission_id	Long	FK → TASK_SUBMISSION	NOT NULL	Identifier of the submission
file_name	String		NOT NULL	File name
mime_type	String		NOT NULL	Mime type
size_bytes	Long		NOT NULL	Attachment size in bytes
file_ref	String		NOT NULL	Reference to the stored attachment

TASK_REVIEW | Room: task_reviews

Field	Type	Key	Constraints	Description
review_id	Long	PK	NOT NULL; auto-generated	Identifier of the review
submission_id	Long	FK → TASK_SUBMISSION	NOT NULL	Identifier of the submission
reviewer_id	Long	FK → USER	NOT NULL	Identifier of the reviewer
review_stage	String		NOT NULL	Review stage
decision	String		NOT NULL	Decision
comment	String?		NULL allowed	Comment
reviewed_at	Long		NOT NULL	Timestamp for reviewed

CONVERSATION | Room: conversations

Field	Type	Key	Constraints	Description
conversation_id	Long	PK	NOT NULL; auto-generated	Identifier of the conversation
created_at	Long		NOT NULL	Timestamp for created

CONVERSATION_MEMBER | Room: conversation_members

Field	Type	Key	Constraints	Description
conversation_id	Long	PK + FK → CONVERSATION	NOT NULL; composite PK	Identifier of the conversation
user_id	Long	PK + FK → USER	NOT NULL; composite PK	Identifier of the user
last_read_at	Long?		NULL allowed	Time this member last read the conversation

MESSAGE | Room: messages

Field	Type	Key	Constraints	Description
message_id	Long	PK	NOT NULL; auto-generated	Identifier of the message
conversation_id	Long	FK → CONVERSATION	NOT NULL	Identifier of the conversation
sender_user_id	Long	FK → USER	NOT NULL	Identifier of the sender user
body	String?		NULL allowed	Message or notification text
sent_at	Long		NOT NULL	Timestamp for sent

MESSAGE_ATTACHMENT | Room: message_attachments

Field	Type	Key	Constraints	Description
attachment_id	Long	PK	NOT NULL; auto-generated	Identifier of the attachment
message_id	Long	FK → MESSAGE	NOT NULL	Identifier of the message
file_name	String		NOT NULL	File name
mime_type	String		NOT NULL	Mime type
size_bytes	Long		NOT NULL	Attachment size in bytes
file_ref	String		NOT NULL	Reference to the stored attachment

NOTIFICATION | Room: notifications

Field	Type	Key	Constraints	Description
notification_id	Long	PK	NOT NULL; auto-generated	Identifier of the notification
recipient_id	Long	FK → USER	NOT NULL	Identifier of the recipient
event_type	String		NOT NULL	Event type
body	String		NOT NULL	Message or notification text
created_at	Long		NOT NULL	Timestamp for created
read_at	Long?		NULL allowed	Timestamp for read

Field	Type	Key	Constraints	Description
task_id	Long?	FK → TASK	NULL allowed	Identifier of the task
week_id	Long?	FK → TRAINING_WEEK	NULL allowed	Identifier of the week
link_id	Long?	FK → SUPERVISOR_LINK	NULL allowed	Identifier of the link
conversation_id	Long?	FK → CONVERSATION	NULL allowed	Identifier of the conversation

4. Relationships

Relationship	Type	Implemented by (FK or junction table)	On delete
USER → TRAINING_RECORD	1 : 0..N	student_id → USER.user_id	CASCADE
TRAINING_RECORD → SUPERVISOR_LINK	1 : 0..N	record_id → TRAINING_RECORD.record_id	CASCADE
USER → SUPERVISOR_LINK	1 : 0..N	supervisor_id → USER.user_id	CASCADE
TRAINING_RECORD → TRAINING_WEEK	1 : 0..N	record_id → TRAINING_RECORD.record_id	CASCADE
USER → TRAINING_WEEK	0..1 : 0..N	confirmed_by → USER.user_id	SET NULL
TRAINING_WEEK → WEEKLY_EVALUATION	1 : 0..1	week_id → TRAINING_WEEK.week_id	CASCADE
USER → WEEKLY_EVALUATION	1 : 0..N	academic_id → USER.user_id	CASCADE
TRAINING_RECORD → TASK	1 : 0..N	record_id → TRAINING_RECORD.record_id	CASCADE
TASK → TASK_SUBMISSION	1 : 1..N (app rule)	task_id → TASK.task_id	CASCADE
TASK_SUBMISSION → TASK_ATTACHMENT	1 : 0..N	submission_id → TASK_SUBMISSION.submission_id	CASCADE
TASK_SUBMISSION → TASK_REVIEW	1 : 0..N	submission_id → TASK_SUBMISSION.submission_id	CASCADE
USER → TASK_REVIEW	1 : 0..N	reviewer_id → USER.user_id	CASCADE
CONVERSATION → CONVERSATION_MEMBER	1 : 2 (app rule)	conversation_id → CONVERSATION.conversation_id	CASCADE
USER → CONVERSATION_MEMBER	1 : 0..N	user_id → USER.user_id	CASCADE
CONVERSATION → MESSAGE	1 : 0..N	conversation_id → CONVERSATION.conversation_id	CASCADE
USER → MESSAGE	1 : 0..N	sender_user_id → USER.user_id	CASCADE
MESSAGE → MESSAGE_ATTACHMENT	1 : 0..N	message_id → MESSAGE.message_id	CASCADE
USER → NOTIFICATION	1 : 0..N	recipient_id → USER.user_id	CASCADE

Relationship	Type	Implemented by (FK or junction table)	On delete
TASK → NOTIFICATION	0..1 : 0..N	task_id → TASK.task_id	SET NULL
TRAINING_WEEK → NOTIFICATION	0..1 : 0..N	week_id → TRAINING_WEEK.week_id	SET NULL
SUPERVISOR_LINK → NOTIFICATION	0..1 : 0..N	link_id → SUPERVISOR_LINK.link_id	SET NULL
CONVERSATION → NOTIFICATION	0..1 : 0..N	conversation_id → CONVERSATION.conversation_id	SET NULL

5. Screen-to-Data Map

This map describes intended Room access by screen. As of main commit e21faf3 (PR #13), some screens are placeholders or accept UI data without a Room binding. DAO availability does not indicate that a screen calls it.

Screen	Reads (tables / fields)	Writes (insert / update / delete)
SplashScreen / WelcomeScreen	— (static onboarding)	—
LoginScreen	USER: email, password_hash, password_salt, role	—; session state is separate
CreateAccountScreen	—; not included in the current app; users log in with preconfigured accounts	—
ResetPasswordSheet (future scope)	—; UI accepts email input	No Room write. Reset email delivery is not implemented.
StudentHomeScreen (planned)	TRAINING_RECORD, TRAINING_WEEK, TASK, TASK_SUBMISSION, TASK_REVIEW, NOTIFICATION	—
TasksScreen / TaskDetailsScreen	TASK, TASK_SUBMISSION, TASK_ATTACHMENT, TASK_REVIEW	—
AddEditTaskScreen (add / edit)	TRAINING_RECORD, TASK, TASK_SUBMISSION, TASK_ATTACHMENT	Insert TASK for new task; insert TASK_SUBMISSION version and TASK_ATTACHMENT rows
AddAttachmentSheet	Local file picker metadata	Stage attachment in UI; persist when task is submitted
MyTrainingScreen	TRAINING_RECORD, TRAINING_WEEK, WEEKLY_EVALUATION	Planned: insert TRAINING_WEEK with totals and report reference
WeeklyEvaluationScreen	TRAINING_WEEK, WEEKLY_EVALUATION	—
FieldSupervisorHomeScreen	SUPERVISOR_LINK, USER, TRAINING_RECORD, TRAINING_WEEK, TASK_SUBMISSION, TASK_REVIEW, NOTIFICATION	—
TraineesScreen / StudentsScreen	SUPERVISOR_LINK, TRAINING_RECORD, USER	—; link sheet handles writes
TraineeDetailScreen	USER, TRAINING_RECORD, TRAINING_WEEK, TASK, TASK_SUBMISSION, TASK_REVIEW	Planned: update TRAINING_WEEK confirmed_by and confirmed_at
ReviewTaskScreen / MarkIncompleteDialog	TASK, TASK_SUBMISSION, TASK_ATTACHMENT, TASK_REVIEW	Insert TASK_REVIEW with field decision; may insert NOTIFICATION

Screen	Reads (tables / fields)	Writes (insert / update / delete)
LinkTraineeSheet / LinkStudentSheet	USER: email, role; TRAINING_RECORD; SUPERVISOR_LINK	Insert SUPERVISOR_LINK request; may insert NOTIFICATION
AcademicSupervisorHomeScreen	SUPERVISOR_LINK, USER, TRAINING_RECORD, TRAINING_WEEK, TASK_SUBMISSION, TASK_REVIEW, WEEKLY_EVALUATION, NOTIFICATION	—
StudentDetailScreen	USER, TRAINING_RECORD, TRAINING_WEEK, TASK, TASK_SUBMISSION, TASK_REVIEW, WEEKLY_EVALUATION	—
AcademicWeeklyEvaluationScreen	TRAINING_WEEK, WEEKLY_EVALUATION	Save WEEKLY_EVALUATION grade and comment
ReviewRelevanceScreen	TASK, TASK_SUBMISSION, TASK_ATTACHMENT, TASK_REVIEW	Insert TASK_REVIEW with academic decision; may insert NOTIFICATION
NotificationsScreen (all roles)	NOTIFICATION: recipient_id, event_type, read_at, target IDs	Update NOTIFICATION.read_at
ChatScreen (all roles)	CONVERSATION, CONVERSATION_MEMBER, USER, MESSAGE	Insert CONVERSATION and membership rows when starting a new chat
ConversationScreen (all roles)	CONVERSATION_MEMBER, MESSAGE, MESSAGE_ATTACHMENT, USER	Planned: insert MESSAGE and MESSAGE_ATTACHMENT; call markConversationAsRead on open.
ProfileScreen (planned Room binding)	USER: full_name, email, role, work_organization	—; log out clears session state

6. Room Plan

DAO	Key queries (name + purpose)	Used by (ViewModel / use case)
UserDao	getUserByEmail: login lookup; getUserById: account details; insertUser: persist an account	AuthViewModel; profile and linking use cases
TrainingDao	getTrainingRecordsForStudent: placements; getTrainingWeeksForRecord: weekly history; getEvaluationForWeek: grade; getSupervisorLinksForRecord: assigned supervisors; insertTrainingWeek, insertWeeklyEvaluation, insertSupervisorLink: save records	TrainingViewModel; trainee detail, academic grading and supervisor linking
TaskDao	getTasksForRecord: task list; getSubmissionsForTask: versions; getAttachmentsForSubmission and getReviewsForSubmission: detail; insertTask, insertSubmission, insertAttachment, insertReview: save work and reviews	TaskViewModel; student submission and supervisor review
ChatDao	getConversationsForUser: inbox; getMessagesForConversation: chat history; getAttachmentsForMessage: files; insertConversation, insertConversationMember, insertMessage, insertMessageAttachment: save chat data; updateLastReadAt: store the member's last-read time	ChatViewModel; markConversationAsRead is implemented. Calling it from ConversationScreen is planned.

DAO	Key queries (name + purpose)	Used by (ViewModel / use case)
NotificationDao	getNotificationsForUser: inbox; insertNotification: create alert; markAsRead: store read time	NotificationViewModel; alert creation and read status

Entity classes (Kotlin names)	UserEntity; TrainingRecordEntity; SupervisorLinkEntity; TrainingWeekEntity; WeeklyEvaluationEntity; TaskEntity; TaskSubmissionEntity; TaskAttachmentEntity; TaskReviewEntity; ConversationEntity; ConversationMemberEntity; MessageEntity; MessageAttachmentEntity; NotificationEntity
Type converters / enums needed	No converters needed for current primitive properties. Role, link status, task type, review stage, decision and event type are stored as String. Timestamps are Long.
Sample (seed) data: what and where from	Local demo data from the Android app: student and both supervisor roles. Use fictional records when seeding tasks, training and messages.
Remote source / mock API (if any)	None. The current database is local Room.

7. GitHub Evidence

☑ Files committed to /docs/database: ER diagram image, schema (this PDF or a .md file)

☑ Commit tagged milestone-3-stage-1

☑ All Milestone 3-A issues closed; each member has at least one contribution

☑ Repository is accessible to the instructor

8. AI Use Log

Record any AI help (ChatGPT, Gemini, Claude, Copilot, etc.) that shaped this design. If none was used, write "No AI used". Using AI is allowed; hiding it is not.

Tool	Where used	What it was used for	What we changed or checked ourselves
ChatGPT	ER diagram	Helped draft and refine the entities and relationships.	Compared the diagram with the Room entities and corrected optional links and the message sender relationship.
ChatGPT	Room setup	Explained the Room setup steps using the course material shared on Blackboard.	We used the Blackboard material as the reference and applied the setup steps to our project's Room configuration.
ChatGPT	Room DAO queries	Helped draft selected user lookup queries for student, field supervisor and academic supervisor accounts.	We adapted the suggested user lookup queries to our Room schema and account fields.
Gemini	Screen layouts	Used for screen layout ideas only.	We used the suggestions for screen layouts only; Gemini did not write the database logic.

☑ We can explain every table, key and relationship in this design without help.

☑ The design matches our Milestone 2 architecture and navigation.

☑ Table and field names are the same in the diagram, schema and Room plan.